

Technology Stack

Date: 15 March 2026 **Team ID:** SWTID-2026-7387 **Project Name:** FitBuddy

Maximum Marks: 2 Marks

Define Your Technology Stack

The FitBuddy technology stack identifies the programming languages, frameworks, database, hosting, version control, and third-party tools required to build a responsive and maintainable fitness-support application.

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	React.js, HTML5, CSS3, JavaScript	Provides a responsive and interactive user interface for fitness tracking, workout plans, progress views, and user profile features. React components support reusable UI development.
2	Backend / Server-Side	Node.js with Express.js	Handles server-side business logic, user requests, authentication, workout data, and API communication. Express.js provides a lightweight structure for building REST APIs.
3	Database / Data Storage	MongoDB	Stores user profiles, fitness goals, workout records, progress information, and related application data using a flexible document-based model.
4	Cloud / Hosting / Deployment	Vercel (Frontend) + Render (Backend)	Provides simple cloud deployment for the web application and backend API, with easy updates and scalability suitable for a student project.
5	Version Control & CI/CD	Git + GitHub	Enables team collaboration, source-code version tracking, branching, backup, and controlled integration of project changes.
6	Third-Party APIs / Other Tools	Google Maps API, Firebase Authentication	Google Maps can support location-based fitness activities such as nearby workout locations or walking/running routes. Firebase Authentication can provide secure user sign-in and account management.

FitBuddy Stack Summary: The selected stack is designed to support a responsive fitness application with a reusable frontend, REST-based backend, flexible data storage, cloud deployment, collaborative development, and optional location/authentication integrations.